EA Risk & Position Size Calculator

An expert advisor can open positions without asking you. That is the point of it, and that is the danger. This worksheet works out, in advance, the largest position you are willing to have open — so the EA's own risk setting confirms your decision instead of replacing it.

> **Leveraged trading can lose money faster than you can react.** Most retail accounts that trade leveraged products lose money. Nothing on this sheet predicts a result or guarantees a profit — it exists to make the loss you are accepting a number you chose rather than a surprise.

Step 1 — Decide the money, not the lots

Choose the amount you are prepared to lose on a single trade if the stop is hit. Traders who start from lot size and work backwards almost always risk more than they intended, because the stop distance is decided last and then stretched to fit.

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Risk per trade (money) = Account balance x Risk %
Lots = Risk per trade / (Stop distance in pips x Pip value per lot)
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Worked example — EUR/USD

1. Account balance $10,000. Risk chosen: 1%.
2. Risk per trade = 10,000 x 0.01 = **$100**.
3. The EA's stop sits 20 pips from entry. EUR/USD pip value is **$10 per standard lot**.
4. Lots = 100 / (20 x 10) = **0.50 lots**.
5. Result: if the stop is hit, roughly $100 leaves the account. That is the whole point of the calculation.

Worked example — USD/JPY, where the pip value is not $10

1. Account balance $5,000. Risk chosen: 0.5% = **$25**.
2. USD/JPY at 150.00: pip value = (0.01 / 150.00) x 100,000 = **$6.67 per lot**.
3. Stop distance 25 pips. Lots = 25 / (25 x 6.67) = **0.15 lots**.

Using $10 here instead of $6.67 would have sized the position 50% too large — a mistake that stays invisible until the stop is hit.

Pip value reference — majors and crosses

Pip value depends on the *quote* currency and on your *account* currency. The figures below assume a USD account and a standard lot (100,000 units), and are approximate for any pair where the quote currency is not USD, because the rate moves. Always confirm the live value in your platform before sizing a position.

Pair	Currency quoted	Approx. pip value per standard lot (USD account)	Note
EUR/USD	USD	$10.00	Fixed — USD is the quote currency
GBP/USD	USD	$10.00	Fixed
AUD/USD	USD	$10.00	Fixed
NZD/USD	USD	$10.00	Fixed
USD/JPY	JPY	≈ $6.67	At 150.00 — rises as the rate falls
USD/CHF	CHF	≈ $11.36	At 0.88 — falls as the rate rises
USD/CAD	CAD	≈ $7.35	At 1.36
EUR/GBP	GBP	≈ $12.50	Depends on GBP/USD as well as EUR/GBP

Rule of thumb: when USD is the *quote* currency the pip value is $10 per standard lot. When USD is the *base* currency it is $10 divided by the rate. When neither is USD you need a further conversion, which is most of the reason indices, crypto CFDs and exotic crosses are best sized in the platform's own risk-per-trade field rather than by hand.

Risk per trade — money at risk, by balance

Account balance	Risk 0.25%	Risk 0.50%	Risk 1.00%	Risk 2.00%
$500	$1.25	$2.50	$5	$10
$1,000	$2.50	$5	$10	$20
$2,500	$6.25	$12.50	$25	$50
$5,000	$12.50	$25	$50	$100
$10,000	$25	$50	$100	$200
$25,000	$62.50	$125	$250	$500
$50,000	$125	$250	$500	$1,000

Read the 2% column and then count how many consecutive losses it takes to lose a fifth of the account: ten. A strategy with a 40% win rate will produce runs of eight to ten losses over a long enough sample. That is the arithmetic behind the usual advice to stay at or below 1%.

Why an EA needs a hard cap as well as a percentage

- A percentage risk setting assumes each trade stands alone. A grid or averaging EA does not trade that way — each level adds to a single losing position, so the "risk per trade" is really a slice of the whole basket.

- Set a maximum number of open positions as well as a risk percentage, and decide the equity figure at which you will intervene by hand.
- If the EA lets you cap total drawdown, set it below the level that would make you act emotionally. A limit you will not respect is worse than no limit, because it hides the risk you are actually carrying.

Grid and martingale exposure ladder

Starting lot 0.01, grid step 30 pips, lot multiplier 1.5, EUR/USD at $10 per pip per lot. Levels are added as price moves against the first entry. Figures are unrounded; your broker will round each lot to 0.01, so real exposure is close to but not exactly this.

Level	Lot added	Cumulative lots	Pips adverse	Floating loss	% of a $1,000 account
1	0.010	0.010	0	$0	0.0%
2	0.015	0.025	30	$3	0.3%
3	0.023	0.048	60	$10.50	1.1%
4	0.034	0.081	90	$24.75	2.5%
5	0.051	0.132	120	$49.13	4.9%
6	0.076	0.208	150	$88.69	8.9%
7	0.114	0.322	180	$151.02	15.1%
8	0.171	0.493	210	$247.52	24.8%
9	0.256	0.749	240	$395.30	39.5%
10	0.384	1.133	270	$619.96	62.0%

> Read the last row again. Ten levels of adverse movement — 270 pips, which EUR/USD covers in a slow week — turns a 0.01-lot start into a 1.13-lot position and a 62% loss on a $1,000 account. Nothing about that sequence is unusual. It is the normal way a grid EA fails: the equity curve is smooth for months and then removes the account in an afternoon.

Your own cap — fill this in and keep it

ACCOUNT BALANCE	RISK PER TRADE (%)	RISK PER TRADE (MONEY)
TYPICAL STOP DISTANCE (PIPS)	PIP VALUE PER LOT (YOUR PAIR)	CALCULATED LOT SIZE
MAX SIMULTANEOUS POSITIONS	MAX TOTAL DRAWDOWN TOLERATED	EQUITY AT WHICH YOU INTERVENE

Write the last three in the same place you will look when the position is red and your judgement is worst. A decision made in advance is the only one an EA cannot talk you out of.

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